Table 3: Accuracy (%) in multiclass text classification on Multilingual Amazon Reviews Corpus.

Models	**DE**	**EN**	**ES**	**ZH**	**Avg.**
Zero-shot					
LLaMA-2-7B (zero-shot)	9.23	12.13	7.43	19.65	12.11
Discriminant baselines					
BERT-Base-Multilingual	53.54	53.42	46.22	67.35	55.13
RoBERTa-Base-Multilingual	52.61	52.56	44.32	66.98	54.12
RoBERTa-Large-Multilingual	52.76	53.64	44.76	66.90	54.52
Ours					
LS-LLaMA	56.80	58.82	49.28	68.72	58.41
LS-unLLaMA	**56.90**	**60.20**	**49.68**	**69.70**	**59.21**

that the model may not be suitable for use in other languages." Consequently, we conducted a series of tests to evaluate LLaMA's text classification capabilities under the zero-shot setting and with our proposed label-supervised adaptation method within multilingual environments. In particular, the adopted Amazon Reviews Corpus encompasses 31 product category labels, making the classification more difficult for both LLMs and discriminant baselines. Zero-shot LLaMA-2-7B presents low accuracy when asked to select one label from 31 candidates. With label-supervised finetuning, the model's performance gains improvements ranging from four to sixfold. In particular, compared with the multilingual version of BERT and RoBERTa, our approaches can still gain remarkable improvements of at least 3.36%, 6.56%, 3.46%, and 2.35% on the German, English, Spanish, and Chinese subsets, respectively, underlining the exceptional multilingual learning capacity of LLaMA. These results underscore the superior efficacy of the proposed label-supervised adaptation method when applied to a range of text classification tasks. Furthermore, LS-unLLaMA outperforms LS-LLaMA, suggesting the benefit brought by the removal of the causal mask.

Table 4: F1 score (%) in NER.

Models	**CoNNL2003**	**OntoNotes V5**
Zero- and few-shot		
LLaMA-2-7B (zero-shot)	1.35	1.20
ChatGPT	67.20†	51.10†
GPT-3.5-Turbo	–	18.22‡
Instruction-tuning		
InstructUIE [Wang et al., 2023]	92.94	90.19
Discriminant baselines		
BERT-Base	92.40§	88.88*
BERT-Large	92.80§	89.27
RoBERTa-Base	92.13	91.55
RoBERTa-Large	92.59	91.72
RAN [Li et al., 2023b]	92.68	89.38
Ours		
LS-LLaMA-2-7B	74.76	77.41
LS-LLaMA-2-13B	74.12	77.73
LS-unLLaMA-2-7B	**93.19**	**92.10**
LS-unLLaMA-2-13B	91.46	91.07

† results reported in Li et al. [2023a]
‡ results reported in Wang et al. [2023]
§ results reported in Devlin et al. [2019]
* results reported in Li et al. [2023b]

3.3.3 Named entity recognition

Results on NER tasks were presented in Table 4. The highlight of the results lies in the performance margin between LS-LLaMA and LS-unLLaMA. In text classification tasks, these two variants can produce comparable results. However,

Table 5: Comparison between different pooling strategies

	SST2		SST5	
	LS-LLaMA w/ CM	LS-unLLaMA w/o CM	LS-LLaMA w/ CM	LS-unLLaMA w/o CM
`max`	95.53	**97.36**	58.79	**60.50**
`average`	55.92	95.96	53.80	59.41
`last`	**96.67**	95.76	**62.31**	46.74

CM stands for causal mask.

LS-unLLaMA achieves 18.43% and 14.69% higher F1 scores than LS-LLaMA on CoNNL2003 and OntoNotes, respectively.

The results pertaining to NER tasks are outlined in Table 4. LLaMA-2-7B under zero-shot setting can only produce single-digit result of F1 score, showing that LLaMA does not have suitable knowledge for addressing NER tasks. Our approach presents higher performance than BERT-Large and RoBERTa-Large. What deserves special attention is the distinctive performance margin observed between LS-LLaMA and LS-unLLaMA. In the realm of text classification tasks, these two variants tend to yield comparable results. However, in NER tasks, LS-unLLaMA exhibits much more reliable performance. Specifically, LS-unLLaMA achieves F1 scores that are 18.43% and 14.69% higher than those achieved by LS-LLaMA on the CoNLL2003 and OntoNotes datasets, respectively. This performance gap indicates the pivotal of global dependencies, encoded within token representations and facilitated by bidirectional self-attention mechanisms, in effectively addressing NER challenges. Such features were absent in LS-LLaMA and other LLMs.

3.4 Ablation study

3.4.1 Pooling method

We conducted a comprehensive study of various pooling methods, including `max`, `average`, and `last`, on the SST2 and SST5 datasets to determine their impact on model performance. The findings, as shown in Table 5, highlight distinct trends. Notably, when employing causal masks, LS-LLaMA achieves the best performance when adopting the `last` pooling method. Conversely, LS-unLLaMA without causal masks demonstrates optimal capabilities when utilizing the `max` pooling strategy.

The reason of such a difference attributes to the presence or absence of causal masks. In LS-LLaMA, causal masks enforces a single-direction information flow, allowing only the last token to attend to all the previos tokens. Consequently, the semantic meaning of the entire sentence aggregates at the last token. In contrast, the tokens within LS-unLLaMA, free from the constraints of causal masks, possess the capacity to attend to every other token in a bidirectional manner. The expanded connectivity facilitates the utilization of max-over-time pooling, which facilitates extracting global dependencies within the sentence.

3.4.2 Training process

We notice that the proposed approaches can outperform BERT on a range of benchmark datasets, hence it is interesting to observe how the LLaMA is finetuned in the same way as BERT. Therefore, we study the training process in 10 epochs of LS-LLaMA, LS-unLLaMA, and BERT on SST5 and ConLL2003 datasets. We recorded the training loss, evaluation loss, and evaluation accuracy / F1 score every 100 training steps and depict the data in Figures 3, 4, and 5. On SST5, in which all these three models performs well, LS-LLaMA and BERT converge faster than LS-unLLaMA, and LS-unLLaMA's testing loss and accuracy tend to be unstable after 5,000 training steps. As discussed in Section 3.3.1, LS-unLLaMA underperforms on small-scale datasets like SST5, as it struggles to reconstruct the self-attention weights that were not well trained during pretraining because of the causal masks. Small datasets cannot provide sufficient label supervision and leads to severer overfitting problem. In contrast, the training process of LS-LLaMA is more stable, and it suffers less overfitting than BERT.

In NER task, the training process of LS-LLaMA is more unstable than those of LS-unLLaMA and BERT, as the token representations of LS-LLaMA intrinsically lacks global dependency due to the causal masks. We can also observe that the testing loss of LS-unLLaMA bounces back later than BERT, suggesting that LS-unLLaMA is more resistant to overfitting issue than BERT with a large training set.

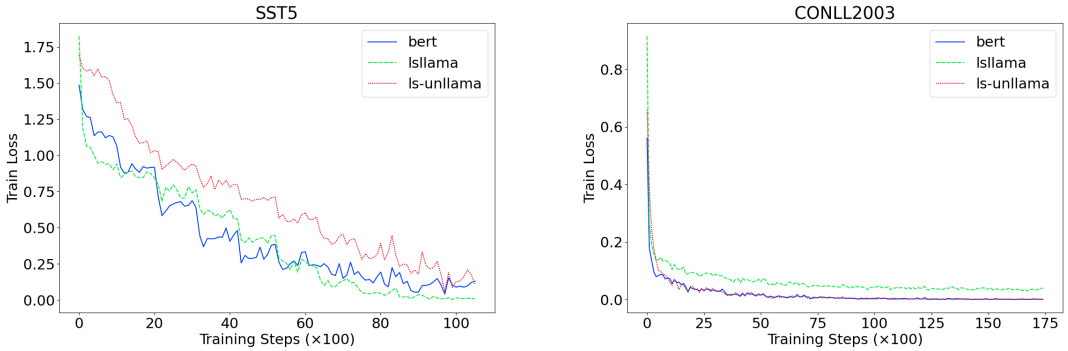

Figure 3: Training loss of BERT, LS-LLaMA, and LS-unLLaMA in 10 training epochs.

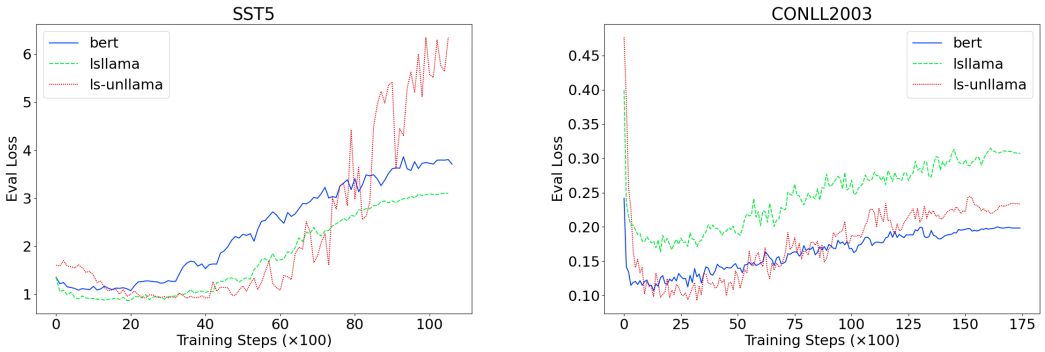

Figure 4: Testing loss of BERT, LS-LLaMA, and LS-unLLaMA in 10 training epochs.

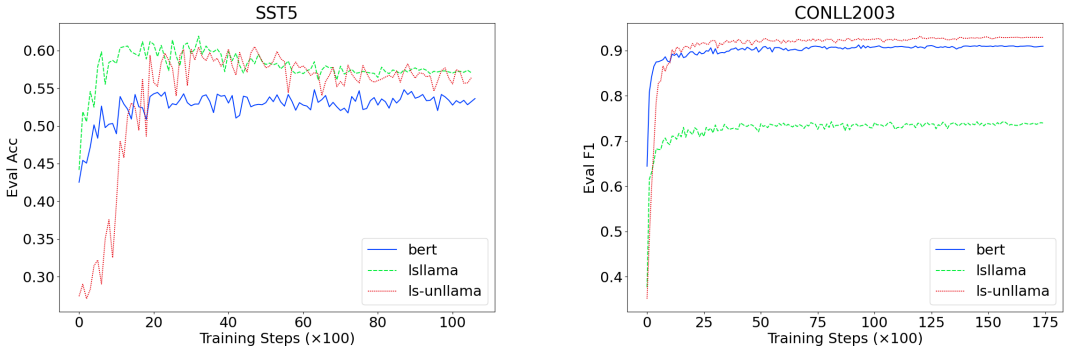

Figure 5: Testing accuracy / F1 score of BERT, LS-LLaMA, and LS-unLLaMA in 10 training epochs.